Progress Report 7

1. **Project title:**
   Touchscreen interface for a multipurpose robot (R1D4) (for the first semester) and swarm of robots to complete locomotive and object-searching tasks (for the second semester).

2. **Names, student numbers, and emails**:
   Group 36 members: (all members are registered in ECSE 458-D1)

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3. **Project advisor(s):**
   Joseph Vybihal: joseph.vybihal@mcgill.ca

4. **Group meetings and meetings with advisor(s):**

All four team members, along with the advisor, have agreed to hold two meetings each week: one with all team members present and another with the advisor and any available team members at that time.

Weekly meeting with the entire group: Wednesdays at 3:00 pm. Weekly meeting with the advisor: Thursdays at 4:30 pm

In addition to these scheduled meetings, this semester, we have decided to dedicate individual free time to working on the swarm of robots at Trottier 4130. This extra lab time will give us the opportunity to work directly with the hardware, identify and fix issues, and improve our system, helping us make steady progress outside of our regular meetings.

1. **Recent progress**

Our recent progress has involved getting the hoverbot up and running on our individual machines and building out our additional computer vision system. In regards to the Hoverbot, it took us a while to get it operating from our machines as there were some issues in the documentation and set up in the git lab. But eventually we got it working with the TCP server and Rest API running on the laptop and the C++ code running on the hover bot. With this set up we were able to send move commands to the hoverbot. On the computer vision side we have created a system that can track the position of the hoverbot from a camera perspective. It uses color to do the tracking. It

measures its position relative to the camera frame and its angle relative to the camera. Additionally, we have also started working on getting the Boebots up and running and adding the new robot.

2. **Future plans**

   We have 3 weeks left until the due date of the Final Report, April 11th. We have set milestones for each week until then as follows:

- 23rd March: Have the basic version of the navigation system running on the Hoverbot as well as be able to send commands from the server to the Boebots.
- 24th - 30th March: Fine tune the navigation system and make it as accurate as possible , contact the R1D4 team to get updates on their system and setup the new Boebot including its hardware and be able to send it commands as well
- 31st March - 6th April: Make final changes if required, adapt the app to match the current configuration of the R1D4, document the code and add documentation for all our changes and finally work on the poster for Design Day. Additionally, start working on the Final Report.
- 7th - 8th April - Prepare for Design Day and practice our presentation.
- 9th - 11th April - Work on the Final Report and submit all the code as four .zip files to our project adviser along with the Final Report. That's the end of the project!